1. Score scale and vote agreement

1. Raw score and normalized margin

Each of T binary classifiers h_t carries vote weight α_t. Its output and the label y are $+1$ or -1. Write

$$F_T(x) = \sum_{t=1}^{T} \alpha_t h_t(x), \quad \|\alpha\|_1 = \sum_{t=1}^{T} |\alpha_t|.$$

The **signed score** is $s = yF_T(x)$. For a nonzero coefficient vector, the **normalized margin** is

$$\rho(x, y) = \frac{s}{\|\alpha\|_1} \in [-1, 1].$$

A positive margin gives the correct label and a negative margin gives an error. Predict $+1$ at a score tie. Zero coefficients throughout make the margin undefined.

2. What a common scale changes

Multiply every coefficient by the same $c > 0$. The signed score becomes cs and its denominator becomes $c\|\alpha\|_1$. The margin and all predicted labels stay fixed. The row loss changes from e^{-s} to e^{-cs}.

If all signed scores are positive, taking $c > 1$ lowers every loss with the same correct predictions. A change in the **relative** vote weights can change margins even when the training error stays zero.

2. Read the training margins

1. Vote shares and the convex hull

For nonnegative coefficients with $A = \sum_t \alpha_t > 0$, let $\lambda_t = \frac{\alpha_t}{A}$. The normalized score

$$f(x) = \frac{F_T(x)}{A} = \sum_t \lambda_t h_t(x)$$

is a convex average: $\lambda_t \geq 0$ and $\sum_t \lambda_t = 1$. The convex hull of a base class $\mathcal{H}$ contains all such finite averages with $h_t \in \mathcal{H}$. The margin is $yf(x) = 1 - 2q$, where q is the vote share on the wrong label. Signed coefficients require the corresponding negated functions to express the same normalized vote this way.

2. Low-margin fraction

For row i of m training rows let $\rho_i = \rho(x_i, y_i)$. At threshold θ, the fraction at or below it is

$$\hat{\rho}_\theta = \frac{1}{m} \sum_{i=1}^{m} \mathbb{1}[\rho_i \leq \theta].$$

The indicator $\mathbb{1}$ counts ties at the threshold. This step function is nondecreasing. At zero it counts all nonpositive margins, including correct ties.

Four rows. Vote A has margins $(0.1, 0.5, 0.6, 0.8)$ and B has $(0.4, 0.5, 0.5, 0.6)$. Both get every label right.

Threshold	A	B
0.20	$\frac{1}{4}$	0
0.50	$\frac{2}{4}$	$\frac{3}{4}$

B has the larger minimum. Its advantage in low-margin fraction depends on the threshold.

3. Bound population error

1. Uniform margin theorem

Fix a binary base class $\mathcal{H}$ **before sampling**, with VC dimension $d \geq 1$. Draw $m \geq 2d$ independent labeled examples from the same distribution $\mathcal{D}$. Choose $\delta \in (0, 1)$.

With probability at least $1 - \delta$ over the sample, the following holds **simultaneously** for every $f \in \operatorname{conv}(\mathcal{H})$ and every $\theta \in (0, 1]$:

$$R_{\mathcal{D}}(h_f) \leq \Pr_{(x,y)\sim\mathcal{D}}(yf(x) \leq 0)$$

$$\leq \hat{\rho}_\theta + C\sqrt{\frac{d \log^2\left(\frac{m}{d}\right)}{m\theta^2} + \frac{\log\left(\frac{1}{\delta}\right)}{m}}.$$

Here h_f predicts from the sign of f with $+1$ at zero. $R_{\mathcal{D}}(h_f)$ is its error probability on a fresh labeled draw. $C > 0$ is a universal, unspecified constant. Here $\rho_i = y_i f(x_i)$.

2. Using the two terms

Increasing θ can count more rows but reduces the complexity penalty. At the same m, d, δ, θ, two votes share a penalty, so the low-margin fractions give the ordering of their bounds. Actual risks can have either ordering.

The same event covers a vote or threshold selected from the sample. It assumes the fixed class used to determine d. The number of rounds enters through the fitted margins, with no explicit round count in the penalty.

4. Compare coordinate directions

1. One coefficient per base function

For a fixed dictionary $\mathcal{H} = \{h_1, ..., h_N\}$, coefficient β_j multiplies base function h_j. The raw score and mean exponential loss are

$$F_\beta(x) = \sum_{j=1}^{N} \beta_j h_j(x),$$

$$J(\beta) = \frac{1}{m} \sum_{i=1}^{m} e^{-y_i F_\beta(x_i)}.$$

Adding ah_j to the score changes only coefficient j by a. Compare all candidates at this same score, using its normalized row losses:

$$s_i = y_i F_\beta(x_i), \quad D(i) = \frac{e^{-s_i}}{\sum_{k=1}^{m} e^{-s_k}}.$$

2. Weighted error gives the slope

Let ε_j be the total D-mass on rows where h_j is wrong. The D-mean of $y_i h_j(x_i)$ is $1 - 2\varepsilon_j$. Differentiating with respect to coefficient j gives

$$\frac{\partial J}{\partial \beta_j} = -J(\beta)(1 - 2\varepsilon_j).$$

Error below one half gives descent for a small positive step. The smaller the error, the steeper the descent among **positive additions** of binary functions.

For signed directions, compare absolute slopes. If two errors are 0.40 and 0.90, their slopes divided by J are -0.20 and $+0.80$. Decreasing the second coefficient descends faster than increasing the first.

5. Select a direction and step length

1. When AdaBoost is greedy

Use a **fixed, finite binary dictionary closed under negation**: if h is allowed, so is $-h$. Their errors sum to one. An exact search for the smallest error under the current D therefore chooses the descending sign of a coordinate with maximal absolute slope when the best error is below one half. Resolve ties consistently.

A learner that returns any positive-edge classifier gives a descent direction. The greedy claim requires exact selection among all candidates.

2. Exact line search on the chosen coordinate

Let e_j be the unit vector in coordinate j and ε the selected function's weighted error. Holding all other coefficients fixed gives the ratio

$$\frac{J(\beta + ae_j)}{J(\beta)} = (1 - \varepsilon)e^{-a} + \varepsilon e^{a}.$$

For $0 < \varepsilon < \frac{1}{2}$ its unique minimum occurs at

$$a = \frac{1}{2} \log \frac{1 - \varepsilon}{\varepsilon}.$$

Exact selection and this line search give one greedy signed-coordinate descent step. At $\varepsilon = \frac{1}{2}$ the best step is zero. At $\varepsilon = 0$ the ratio e^{-a} has no finite minimum; return the perfect training classifier. A line minimum need not minimize the full objective.

6. What one step establishes

1. A loss decrease can lower a margin

In the four-row example, classifier h_j misses only row j. Start at vote A with coefficients $(0.45, 0.25, 0.20, 0.10)$. Include both signs of each h_j in the search. The best candidate is h_4, with error about 0.17905 and step $a \approx 0.76140$.

Quantity	Before	After
Mean loss J	0.62738	0.48107
Minimum margin	0.10000	0.02192
Training mistakes	0	0

Adding h_4 helps rows 1–3 and hurts row 4. Mean loss falls while the minimum normalized margin also falls.

2. Why a hard row can dominate

For two rows with signed scores s_i, s_k,

$$\frac{D(i)}{D(k)} = e^{s_k - s_i}.$$

Scores -1.2 and 1.6 give a mass ratio of $e^{2.8} \approx 16.44$. Doubling both raises it to $e^{5.6} \approx 270.43$ with the same normalized margins under common coefficient scaling. A large mass can reflect a difficult row, a wrong label or a restricted base class.

3. Evidence for stopping

Use validation data to choose a round. Evaluate the chosen procedure on separate test data or by outer cross-validation.